

Automated Bingo Card Generation

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Summary

My 'Bingo' repository contains two main files: **bingo_75.py** and **bingo_MM.py**. Through both, you can generate an adjustable number of unique, playable bingo cards, and their values write into a CSV file in wide format.

Combine a design with inputted cell values, and you have bingo cards!
Mass generated !!! :]

This project has 2 versions .

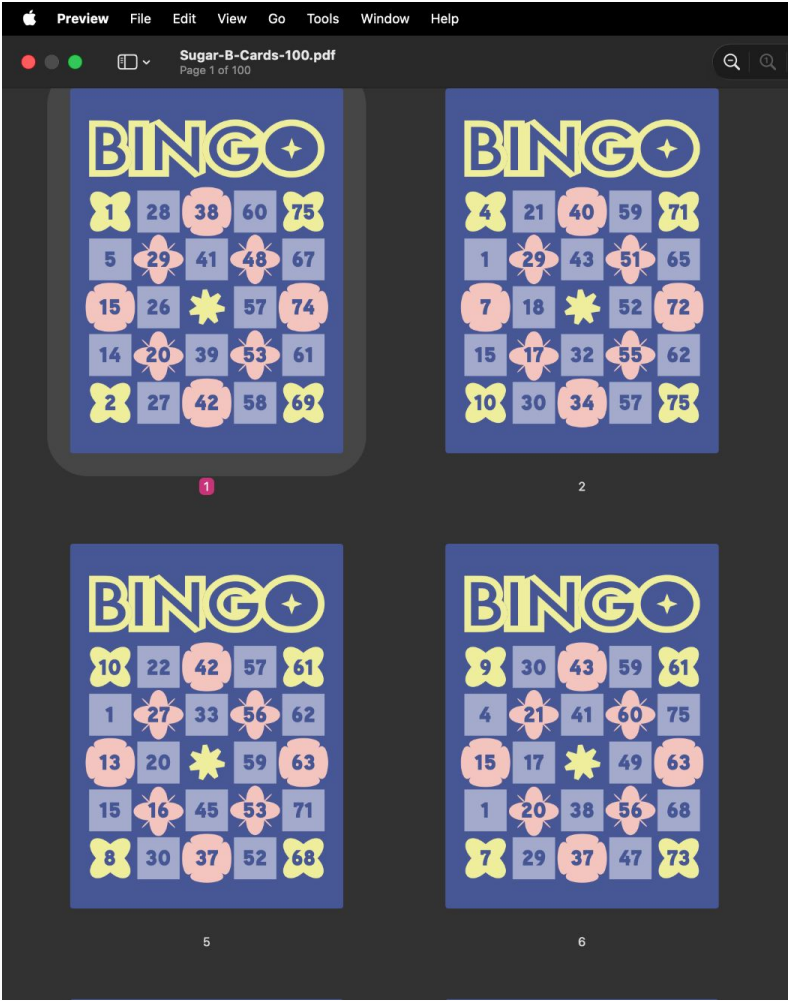
Basic 75-Ball Bingo

- 75-Ball Bingo is the most popular version of Bingo in the United States.
- It is made up of 5 columns and 5 rows.
- Each column corresponds to a number range: B (1–15), I (16–30), N (31–45), G (46–60), and O (61–75), with a “FREE” center space.
- All cards are random and unique.

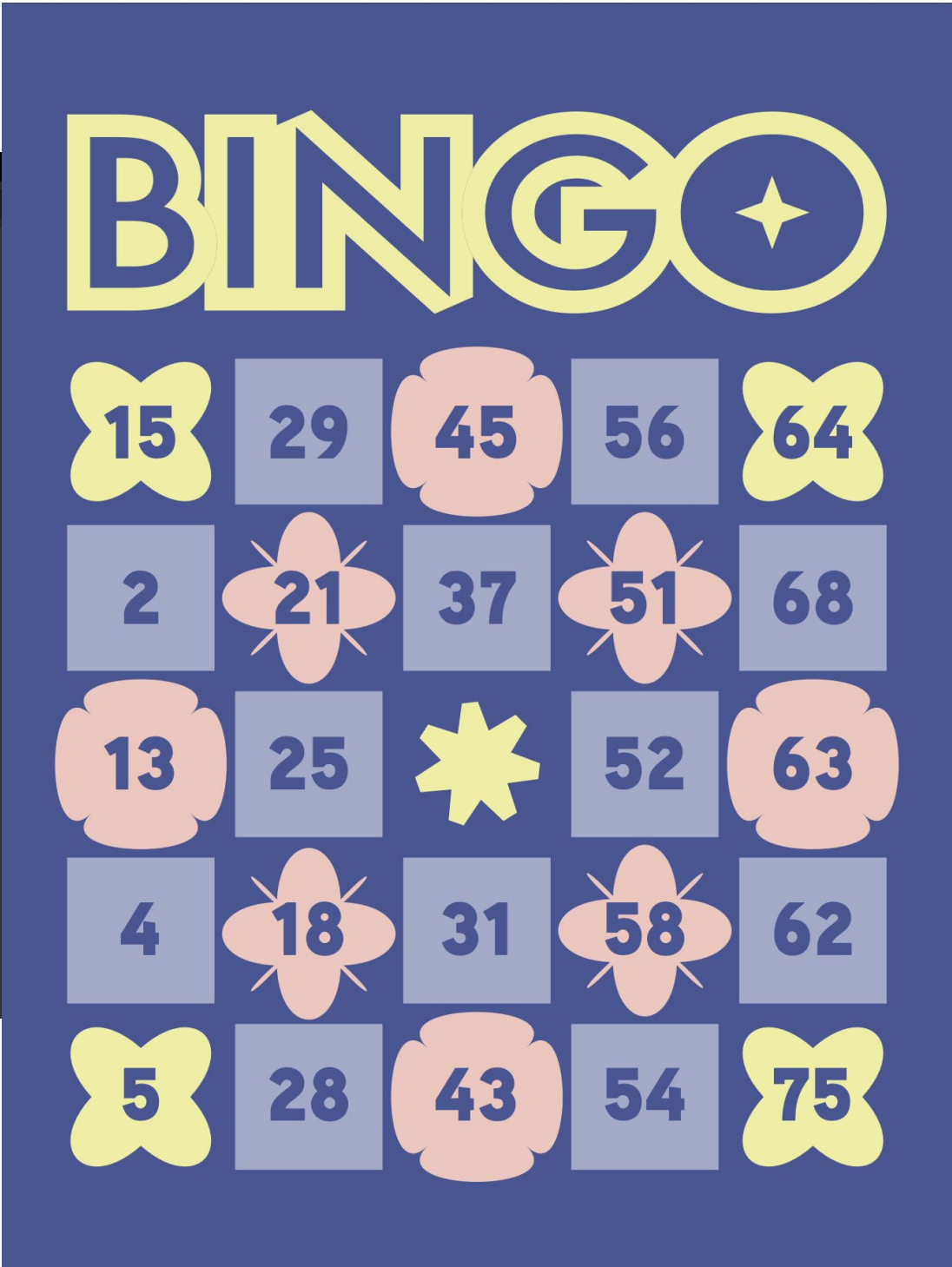
My Book Marketing Idea

- Also made up of 5 columns and 5 rows, except these do **NOT** contain number values.
- Rather, they contain **True/False statements** I defined in a CSV.
- While all cards are *also* random and unique, there the number of winning and losing cards is an **editable** parameter.

My Results !!



75-Bingo



WINNER!

Tools & Softwares

02



Excel

to create a content bank of
true / false squares
(for Part 2 of project)



Python

to create randomized 5x5
game matrices



Illustrator

to design bingo cards on-theme
to my novel



InDesign

to populate the squares of all
bingo cards

Program Structure

My 'Bingo' repository contains two main files: **bingo_75.py** and **bingo_MM.py**. Through both, you can generate an adjustable number of unique, playable bingo cards, and their values write into a CSV file in wide format.

This next slide will only cover one part of this project: **bingo_MM.py**.

The script reads TRUE and FALSE statements from a CSV file (a template is provided).

In a single function, `create_set()`, you can generate a custom number of Bingo cards, specifying how many should be winning cards and how many should be losing cards.

This function utilizes the `winning_patterns`, `truth_statements`, and `false_statements` lists, along with previously defined functions: `count_winning_patterns()`, which ensures losing cards contain no winning patterns, and `create_winning_stack()` and `create_losing_stack()`, which generate the respective card types.

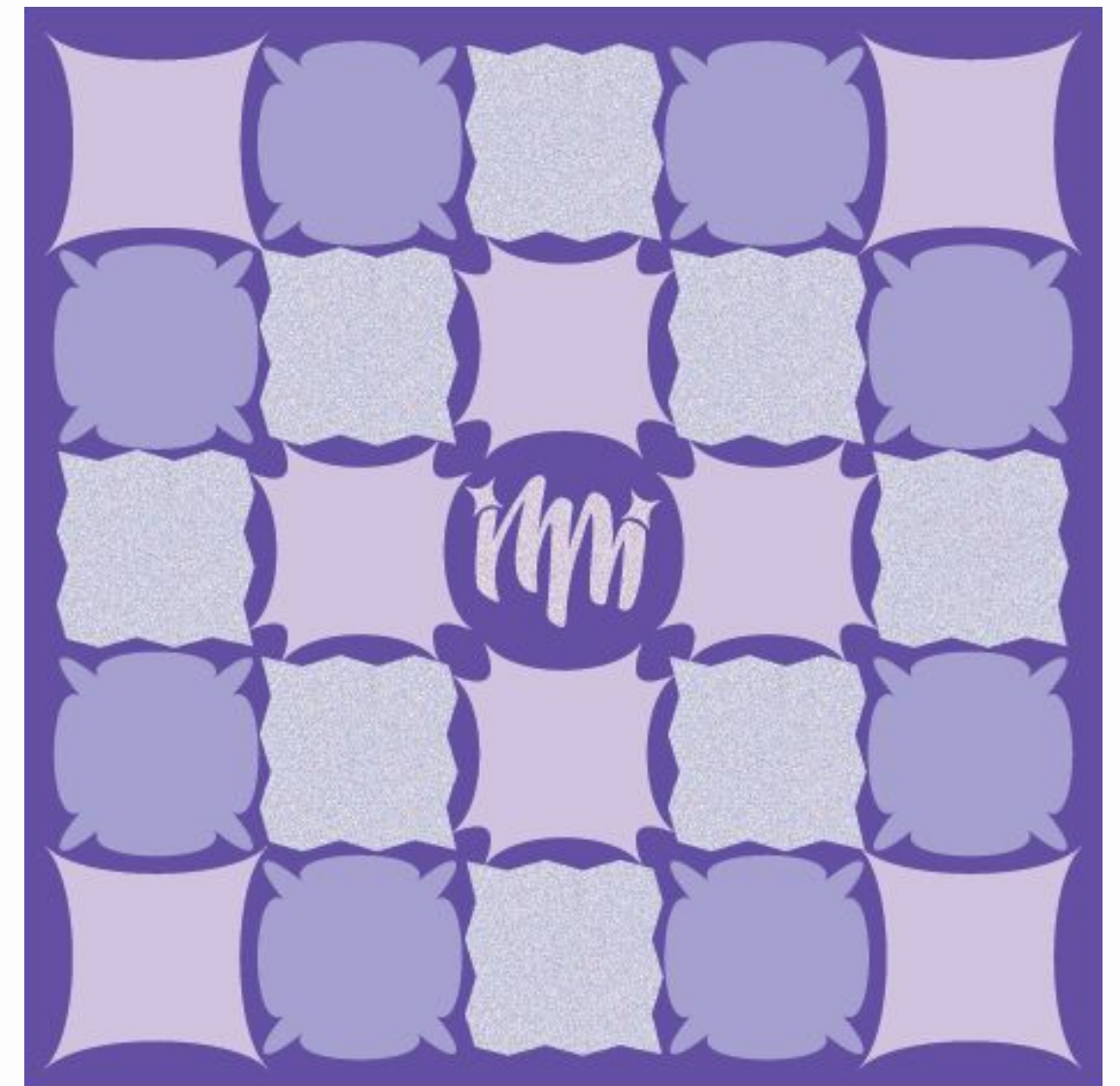
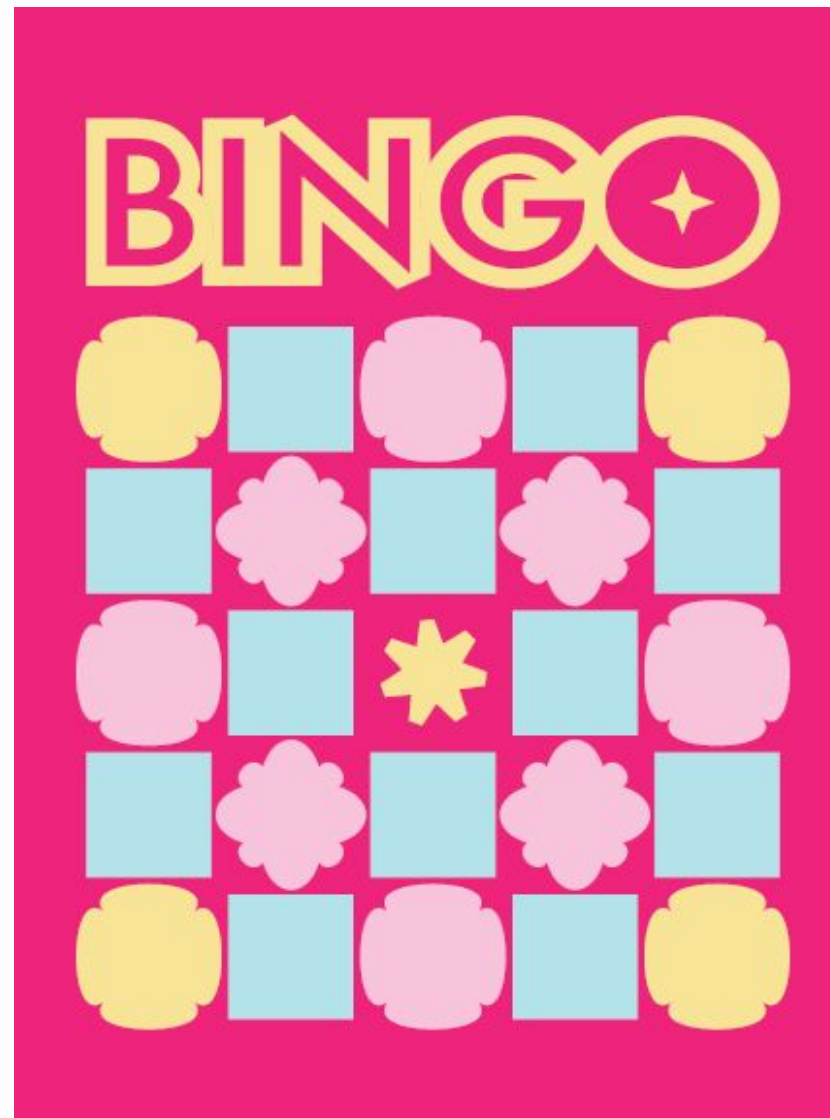
```
1  import pandas as pd
2  import random
3
4  # Bingo Card Arrangement
5  # -----
6  # 00  01  02  03  04      B
7  # 05  06  07  08  09      I
8  # 10  11  12  13  14      N
9  # 15  16  17  18  19      G
10 # 20  21  22  23  24      O
11
12 > winning_patterns = [...
33
34 # Using the "Bingo Card - T_F Bank.csv" file, this code reads the TRUE and FALSE
   statements into lists. If your file goes by a different name, you will need to change
   the file name in the code below. The file should be in the same directory as this
   script as well.
35 truth_statements = pd.read_csv("Bingo Card - T_F Bank.csv")["TRUE"].dropna().tolist()
36 false_statements = pd.read_csv("Bingo Card - T_F Bank.csv")["FALSE"].dropna().tolist()
37
38 > bingo_columns = [...
45
46 # VALIDATION FUNCTION
47 > def count_winning_patterns(card): ...
57
58 # CARD GENERATOR FUNCTIONS
59 > def create_winning_stack(total_winning_cards): ...
114 > def create_losing_stack(total_losing_cards): ...
158
159 # MAIN FUNCTION (CREATES SET)
160 > def create_set(total_winning_cards = 1, total_losing_cards = 10): ...
181
182 # create_set(REPLACE_WITH_COUNT_OF_WINNING_CARDS, REPLACE_WITH_COUNT_OF_LOSING_CARDS)
183
184 # Example usage:
185 your_set = create_set(3, 7) # This will create a set of 3 winning cards and 7 losing
   cards, for a total of 10 cards. You can change these numbers out.
186 your_set.to_csv("your_set.csv", index=False)
```

Design Choices

Jokes! There is no strategy going on here. Fun shapes and poppy colors are just my vibe :)

Made using Adobe Illustrator.

- Shapes
- Effects
- Text



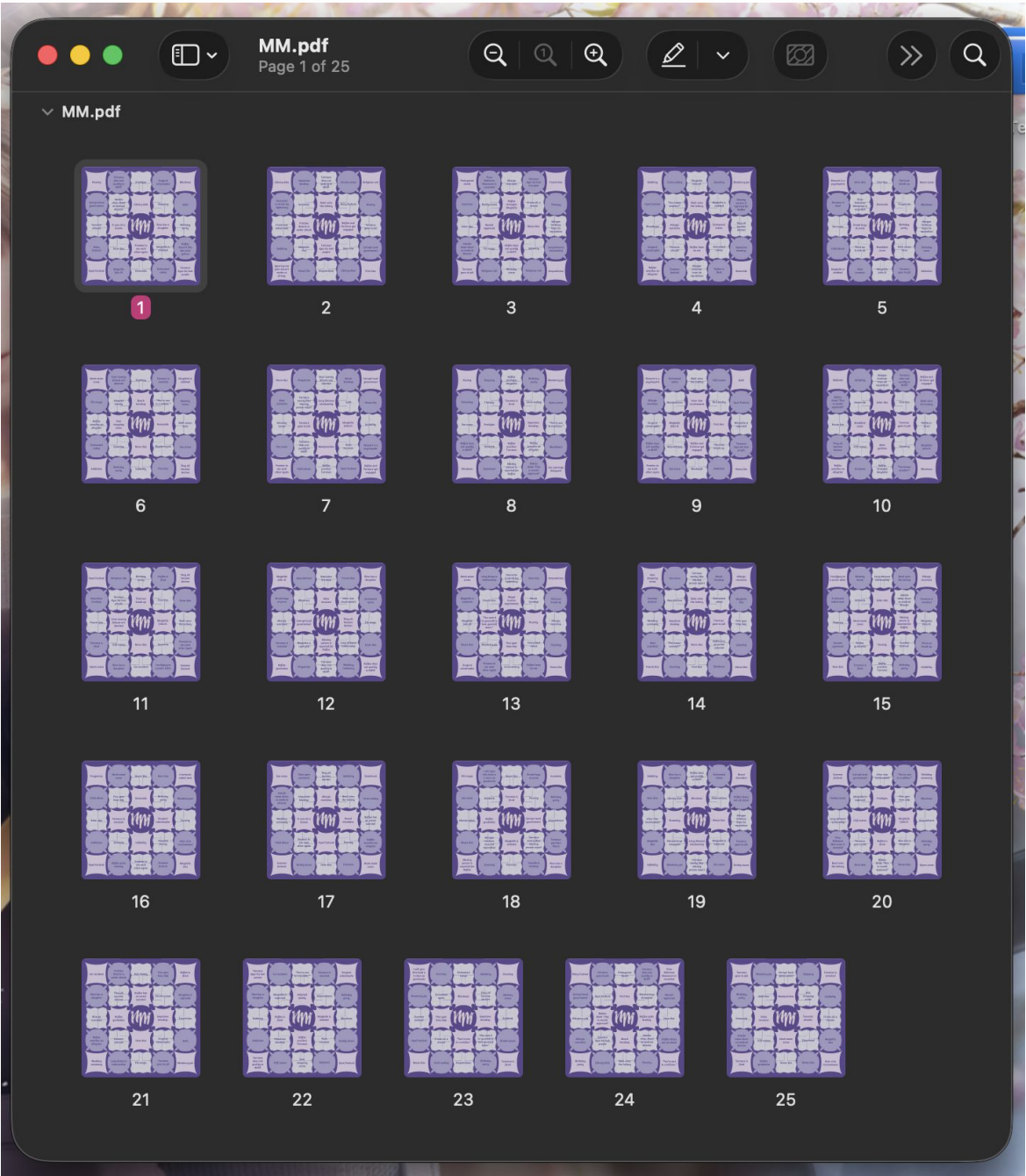
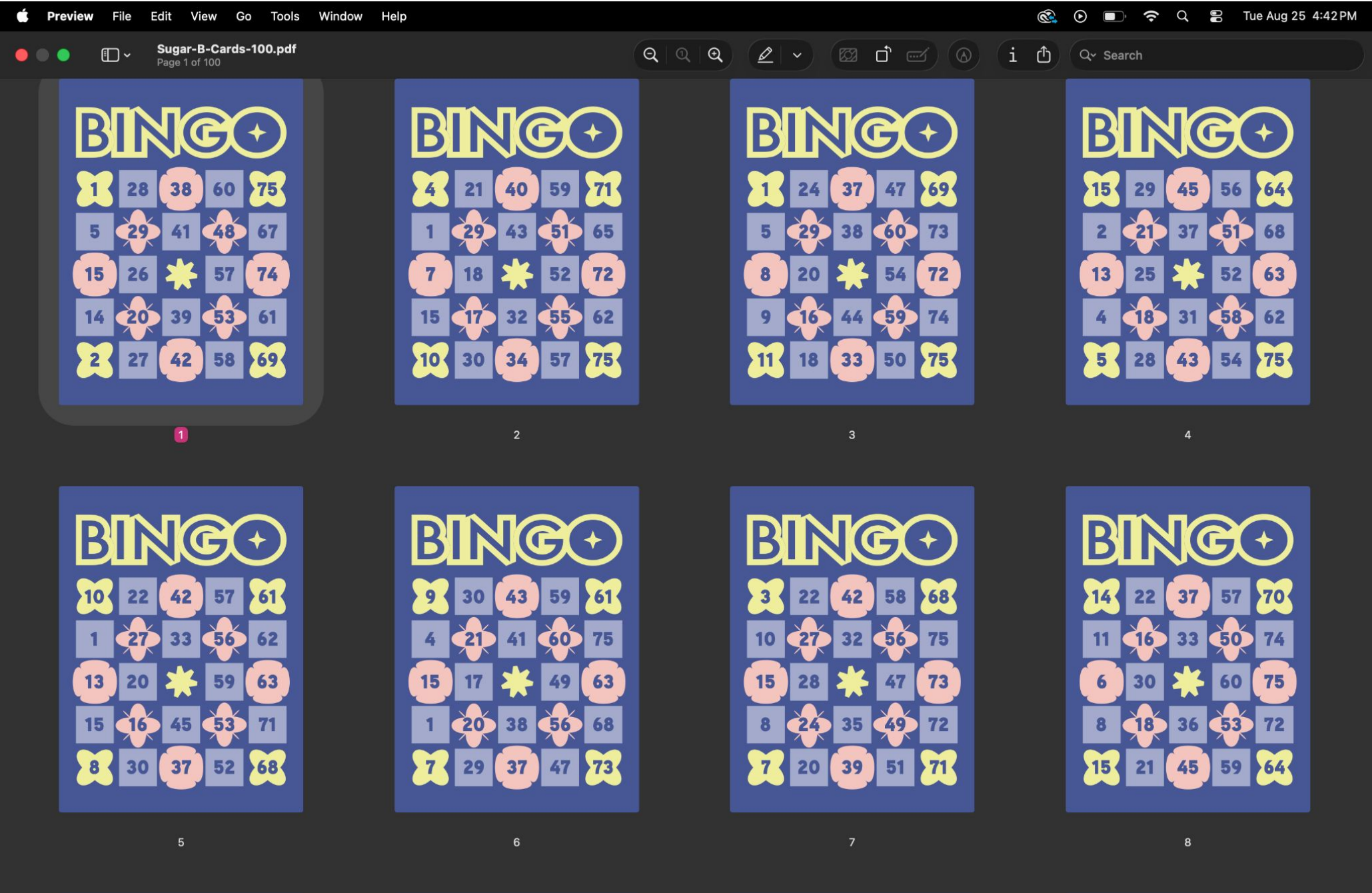
Generating Cards

STEPS SIMPLIFIED:

1. Clone my repository for python script and functions. Run them.
2. Have a bingo card design/layout.
3. Map bingo values onto the card.
4. Replicate then export.

**I will be using InDesign's Data Merge,
but this is not a tutorial.**

Easily generate as many Bingo Cards as you want!



Conclusion

Bingo_MM.py allows for the mass generation of Bingo cards with a predetermined number of winning and losing cards.

Imagine an author promoting an upcoming book that releases years from now. As part of a pre-order or promotional campaign, readers receive a unique Bingo card. Each cell contains a statement that will eventually be revealed as true or false based on the contents of the book. Readers won't know the answers until the book is released, creating a sense of anticipation and encouraging speculation about what will become canon.

Winning cards could then be rewarded with incentives such as tickets to a related event, exclusive merchandise, or a special edition of the book.



Python Libraries

- Pandas
- Random



Skills Demonstrated

- Conditionals, loops, lists, randomization, functions
- Algorithm design
- Data validation
- File I/O
- Automation